

New fuzzy logic inference rules for optimized reversing and steering control of trailer and truck vehicle

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Abstract— in this paper we present a car parking control for real-time application and also needs less skill for driver. This analysis is applied to control the trailer-truck vehicle with 4 degree of freedom at plate. Fuzzy control technique with different levels of controller and rules is applied to controller for better decision making and control of steering angle. in this article based on the switching takagi-sugeno (T-S) fuzzy control, we design 8 fuzzy controllers for moving in different direction according to the location and distance from parking location. Then the switching T-S fuzzy model is synthesized by a two controller for removing error with considering the desired angle and also control steering angle at any moment and another one for adjusting desired changes the angle of the car in presence of obstacles.

I. INTRODUCTION

Automation is becoming increasingly important role in the global economy and daily experience. And parking system in car and passive trailer system provide very advantages in delivery and transportation systems. However, the major drawback of this system is that the control problems. The trailer system has been studied from nonlinear control theory aspect. For example proof of their controllability...

Nonlinear and complex dynamics of trailer-truck vehicle and changes in their dynamics with both environmental conditions and condition in parking makes instability condition that justifies the use of fuzzy control as shown in Fig. 1. These reasons have caused instead of using a linear controller or Nonlinear Controller or observer, a Fuzzy controller is applied for responding in different situations. Fuzzy logic is derived from two-valued logic. Therefore it is capable of processing data and knowledge on the environment with uncertainty. Uncertainty and ambiguity from input is come to the control system. In 1965 fuzzy set theory is introduced by professor Lotfi Zadeh fuzzy [1] and membership in a fuzzy set is expressed with two right and wrong logic [2]. Due to the non-linear nature and complexity of the model of parking, it is a procedure for investigation of any control method. in parking process the Angle between trailer and truck must be observed and their limitation is considered. in Most controllers the important objective is to prevent to go further trailer for reducing the angle of vehicle.

Many activities on the reverse parking of the trailer-truck have been done. In references [3] and [4] the researchers, have used only one fuzzy controller and then analyzing its stability. It is true that this method is effective, but detailed knowledge

is difficult for non-fuzzy section. In reference [5] Researchers used two fuzzy controller for get ridding of the problem of great angle between the trailer and truck in other words, one is used for going front and another for going back and also Neural networks are well-implemented, but most data needed for training. However, the fuzzy controller needed very small control laws. Very large number of articles can be found for investigation on basis concept of normal driving training [6],[7] and basic concepts of genetic algorithms and neural fuzzy neural network [8],[9],[10] is presented that is shown in fig.2. In the next section, a simple mathematical model of kinematic of vehicle with one truck is presented according to fig.3. fuzzy control with positioning the sensor in simple model is shown in fig.4 and with more detail of controller and experimental setup is shown in fig.5. And in the following sections the reverse parking is solved with sequential fuzzy controller and in the end it is validated by simulation method.

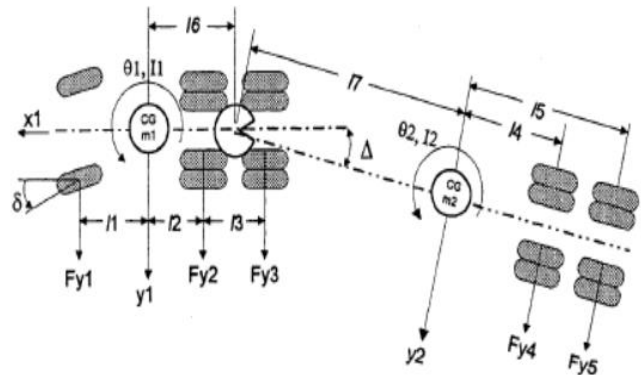


Fig. 1 Dynamic model of the vehicle

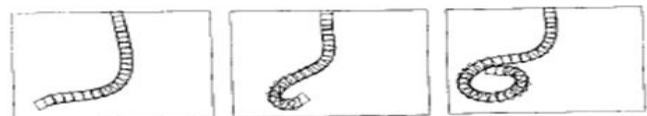
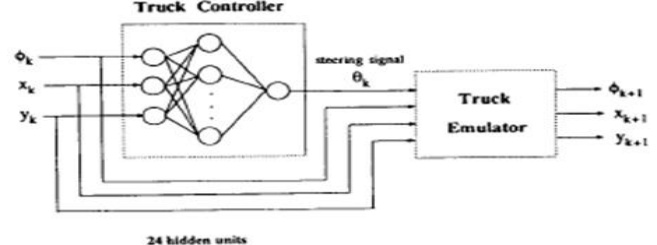


Fig. 2 (1) a block diagram of a neural network, (2) the result of controller

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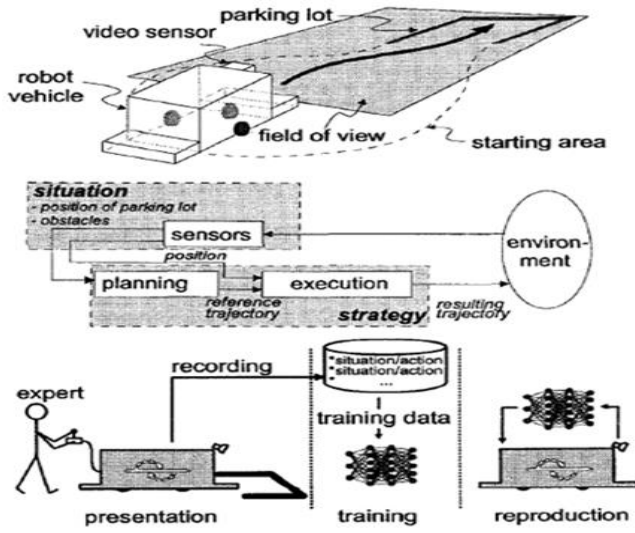


Fig. 3 (1) vehicle schematic and placing the camera, (2) control block diagram (3) The idea of using neural network based control of human behavior.

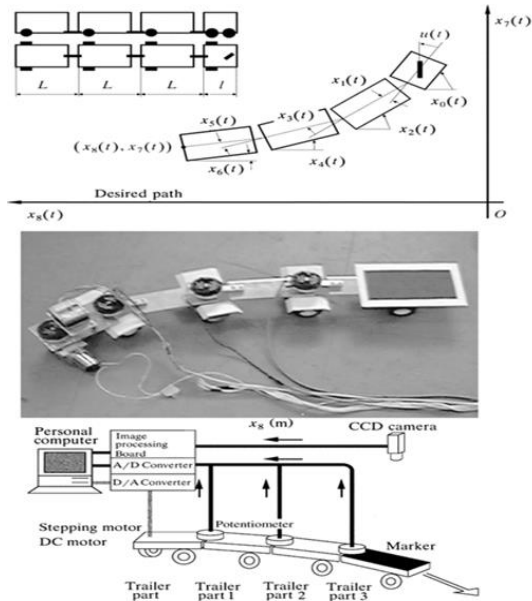


Fig. 4 (1) vehicle kinematic modeling of multiple joints (2) Robots (3) Joint Vehicle Simulator Lab (4) Connections Schematic experimental robot sensors and computer operator

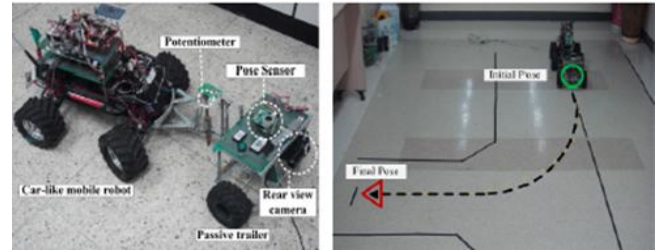
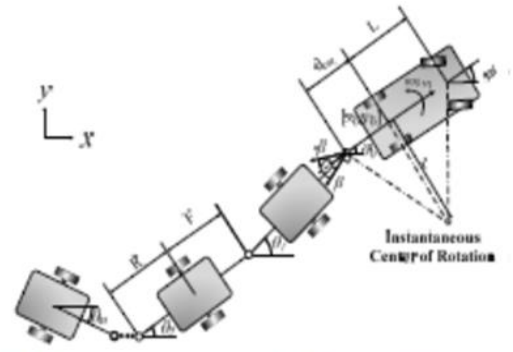


Fig. 5 (1) detailed vehicle kinematic modeling, (2) a detailed scheme of the car park, path, (3) laboratory samples made by the doctor in China

II. DESCRIPTION OF MODELING

A. Introduction of vehicle body components

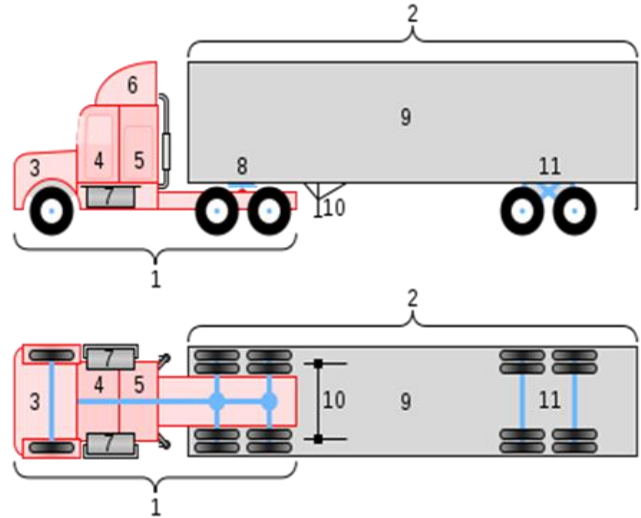


Fig. 6 model of vehicle with different components

The following components are numbered in Fig. 6, respectively, as follows:

- 1) Driving part
- 2) Base truck part
- 3) Axis to connected steering system
- 4) Cabin
- 5) Rest part
- 6) Aerodynamically part
- 7) Fuel tank
- 8) Connection of wheels
- 9) Enclosed loading space
- 10) Separation part
- 11) Back axis of drive

B. Kinematic description of modeling

The basic model of a vehicle with single truck is shown in Fig. 7. The Direction of vehicle is determined by the rotation of the front wheels of vehicle that angle between front wheels with trailer body is defined by α .and angle between Axis angle of trailer with the horizontal axis is defined by β .the Angle between the major axis of truck and the main axis is shown by γ .and Angle between of the major axis and horizontal axis is shown by δ variable. α Angle is indirectly effects on γ . In addition, γ angle directly restricted by the angle between the trailer and truck and in this condition is considered 90 degrees. When the trailer is go back the angle between trailer and truck is finally reached to this limited value and only when the trailer go front this angle will be reduced.in fact the limitation of this angle depend on the maximum angle of front wheels .and length of the trailer is 1 m and truck is 11m .

In References [3] and [4] the simplified mathematical model of the vehicle can be found in (1).

$$\gamma(t) = \beta(t) - \delta(t) \quad (1)$$

$$\beta(t+1) = \beta(t) + \frac{v\Delta t}{l_1} \tan[\alpha(t)] \quad (2)$$

$$\delta(t+1) = \delta(t) + \frac{v\Delta t}{l_2} \sin[\gamma(t)] \quad (3)$$

$$x(t+1) = x(t) + v\Delta t \cos[\gamma(t)] \cos[\delta(t)] \quad (4)$$

$$y(t+1) = y(t) + v\Delta t \cos[\gamma(t)] \sin[\delta(t)] \quad (5)$$

Equation (2) is the variation of angle of trailer which v is the velocity of trailer. From (1-3) we can find the angles that are non-linear dependence. At times $t+1$ we can find the horizontal and vertical position at time t from (4) and (5). These equations are used to evaluate the performance of fuzzy controller and are developed in symbolic of matlab in according to Fig. 8. Velocity is constant and is reached a negative value in reverse parking so model has only one input variable that is called an alpha angle. Using the left block in Fig. 8 the angle of trailer with horizontal axis is achieved according to the (2). The angle of truck in low block is extracted according to (1) and (3). At the top blocks the absolute position of end of the truck is calculated. The angles and with absolute position of end of the truck is used for graphic simulation . Calculation of the angle beta that is done in left block that can be written in format of the below in (6, 7).

$$\frac{\beta(t+1) - \beta(t)}{\Delta t} = \frac{\Delta\beta(t)}{\Delta t} = \frac{v}{l_1} \tan[\alpha(t)]$$

$$\beta(t) = \int_{t_0}^t \frac{v}{l_1} \tan[\alpha(t)] dt + \beta(t_0) \quad (6, 7)$$

Equation (7) represents the angle beta in time t that in this study is used integral format and is shown in Fig. 8 in format of 1/s. And Time $t(0)$ in (7), is the time required to simulate the initial conditions. With the same argument for the horizontal and vertical positions and angles we need the initial value at time t in the integrator block.

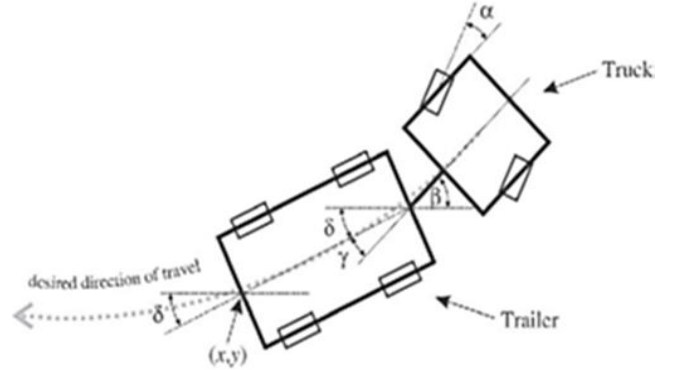


Fig. 7 the basic model of trailer and truck

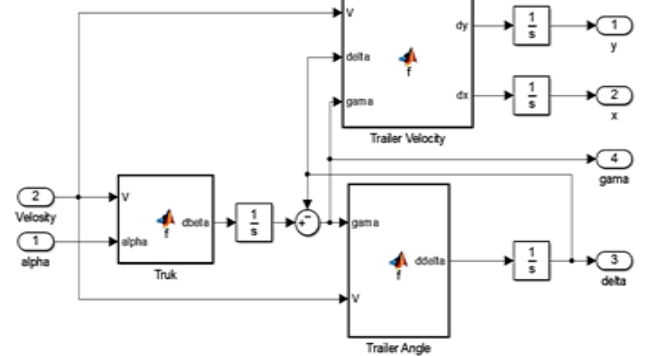


Fig. 8 Block diagram of the kinematic of vehicle

III. DESIGN HIERARCHAL FUZZY SYSTEM CONTROL IN TRAILER-TRUCK VEHICLE

Since the relatively large number of fuzzy rules to control these systems is required so we use the hierarchical fuzzy method consists of simple fuzzy controllers .Processing for the condition of input to system can be done by four distinct stages or sequential or parallel in 3 levels that is shown in Fig. 9. In The first level we use the 8 fuzzy controllers for the decision making to moving in the main or sub-directions. In the second level the takagi sugeno chooser Select the each of the first level results in relative to the distance from the parking lot. Simultaneously in parallel with the first and second stage, the fuzzy controller according to the distance and angle of truck to obstacle is changed the angle of steering. The aggregate result of these two steps in the third level, that is the fuzzy compensator, is the level at which a phase compensator to change the angle of the steering angle to the desired value. In the first level of fuzzy controller that has single-input and single-output. And in the second level of the fuzzy controller it has with 9 inputs and one output. And also the fuzzy

controllers related to obstacle avoidance have with two inputs and one output and finally fuzzy controller in third level have with an input and an output signals.

A. The first level of controller

All eight fuzzy controllers in this level have task for choosing the angle between trailer and truck with respect to the target and the input angle with the horizontal axis that is shown with δ . In fact, these 8 fuzzy controller for moving the angle of truck part at main direction (up-left-right-down) and sub direction (up-right, up-left, down-left, down-right), which can be seen in Fig. 10. In fact, each direction is used in fuzzy format during the movement phase. For example, moving to the left when the trailer vehicle is located the far from the parking lot in right position is used. And this 8 fuzzy controller is shown in Fig. 11. at the proposed time the other controller is activated at around location of parking lot that is moved similar as shown in Figure Fig. 12. The introductions of these fuzzy rules for angle are shown in Fig. 13 - Fig. 20. The inference engine in all of those is mamdani and we used the center of surface for difuzzification. The Sets of fuzzy rules for each are as follows:

If (δ is Very Negative) then (γ is Maximum Left)
 If (δ is Negative) then (γ is Moderate Left)
 If (δ is Suitable) then (γ is Zero)

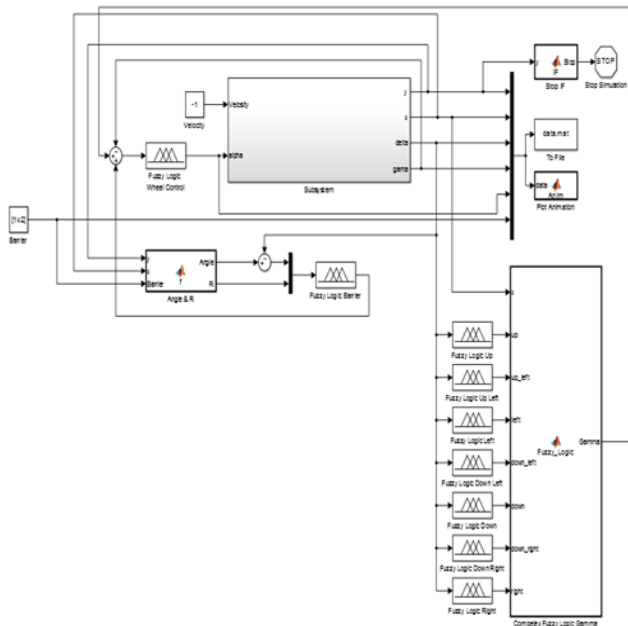


Fig. 9 Block diagram of overall and finalized system

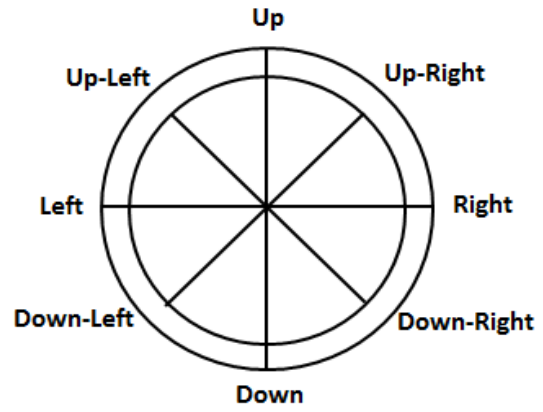


Fig. 10 Division of direction to eight parts
 For example:

If (δ is Positive) then (γ is Moderate Right)
 If (δ is Very Positive) then (γ is Maximum Right)

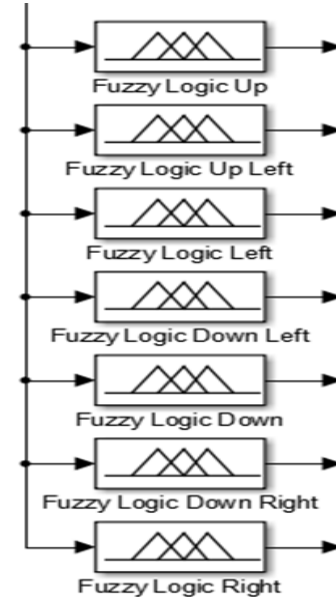


Fig. 11 position of controllers

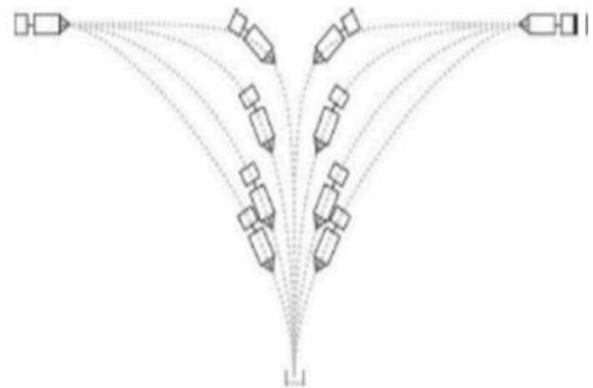


Fig. 12 eight group of fuzzy controller according to time of simulation

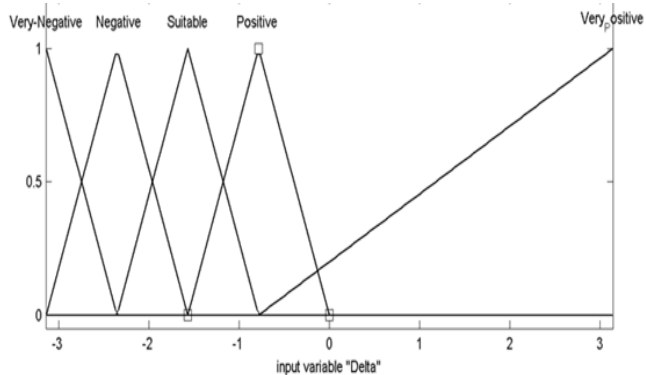


Fig. 13 Introduction to Climbing

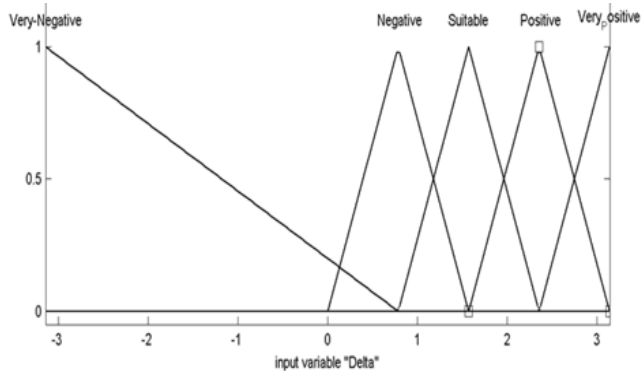


Fig. 17 for bottom

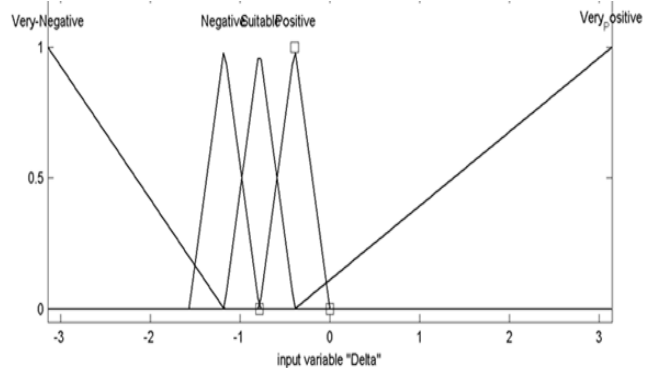


Fig. 14 Top Left

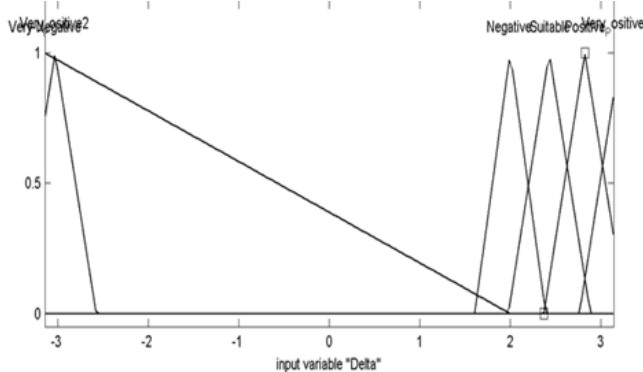


Fig. 18 Introduction Bottom Right

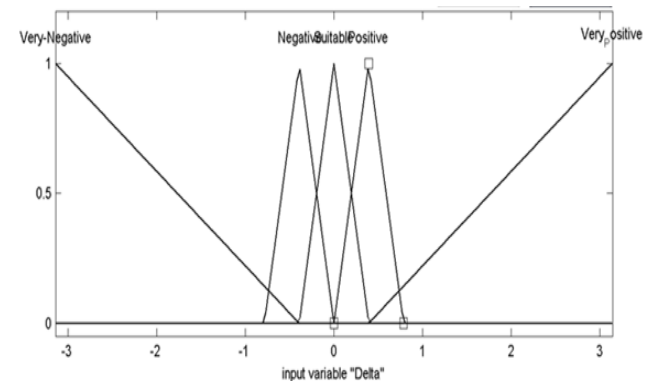


Fig. 15 for left

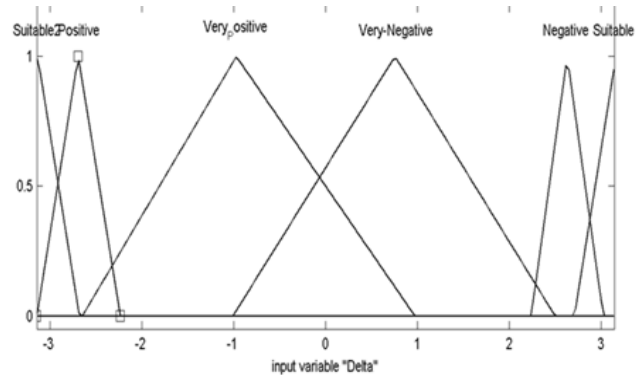


Fig. 19 for right

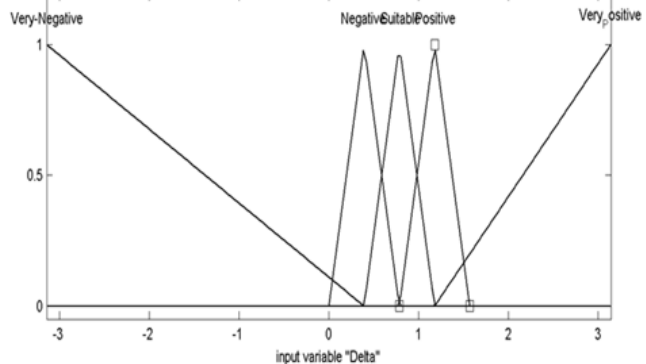


Fig. 16 Introduction Bottom Left

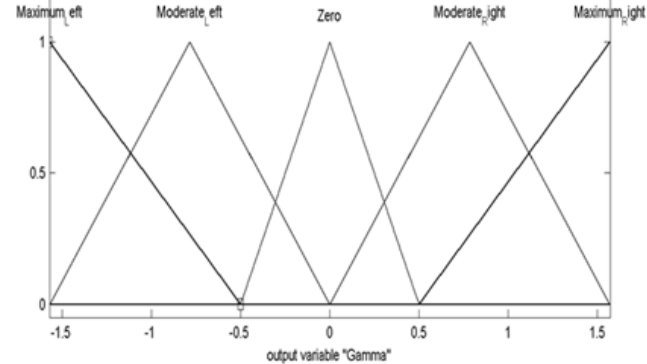


Fig. 20 conclusion

B. The second level of controller

The input to the second level in addition of the 8 output of the fuzzy controller in first level, is the end x position of the trailer. This controller is designed for combines and select fuzzy rules for each truck angles in accordance to the horizontal position. Variable x have five verbal values that is shown in Fig. 21. For defuzzification we use takagi sugeno which sets the rules for each phase is as follows. Note that these rules to be used for the y vertical position of the truck and exact same rules for different are used for negative vertical position.

If (x is Big Left) then (γ is Right)
 If (x is Left) then (γ is Down Right)
 If (x is Zero) then (γ is Down)
 If (x is Right) then (γ is Down Left)
 If (x is Big Right) then (γ is Left)

C. The third level of controller

The objective of this fuzzy controller is for calculating the new angle of the front wheel. The input to it, is the difference between trailer and truck angle from desired second level and the actual angle of gamma. Indeed, as can be seen in Fig. 21 the change angle of the front wheel is not depend to the true angle between the trailer and truck and variation to the angle wheel is caused to increase and decrease of gamma angle. The strategy for computing the new angle is very simple and is as follows. If the actual angle of gamma is more or less than the desired angle then the angle of wheel is changed in the opposite direction. And if the actual angle of gamma is equal to the desired then the angle of wheel does not change. The region of gamma angle is divided to Positive, negative, and zero according to Fig. 22. Similarly the range of the variation of wheel angle is divided to left and right that is shown in Fig. 23. The inference engine in all of those is mamdani and we used the center of surface for difuzzification .the alpha angle is reached from the below fuzzy rules.

If ($\Delta\gamma$ is Negative) then (α is Right)
 If ($\Delta\gamma$ is Zero) then (α is Down Keep)
 If ($\Delta\gamma$ is Positive) then (α is Left)

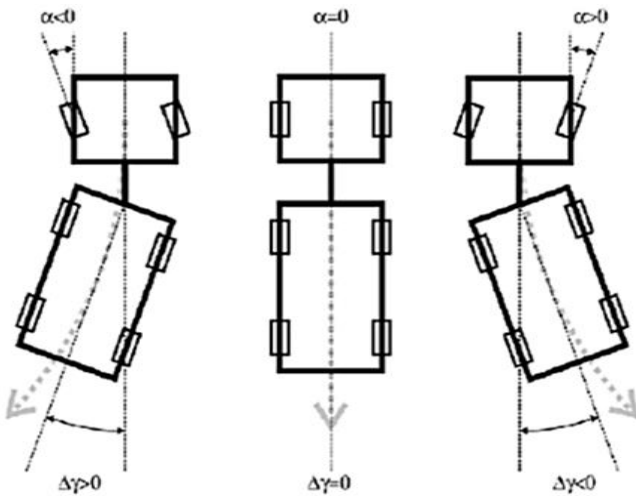


Fig. 21 The change in angle of the vehicle

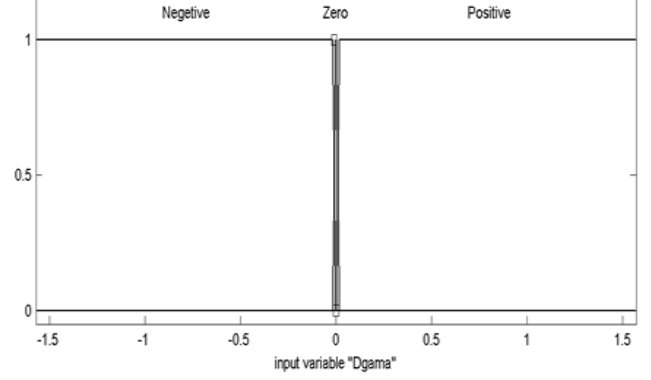


Fig. 22 Introduction to compensation

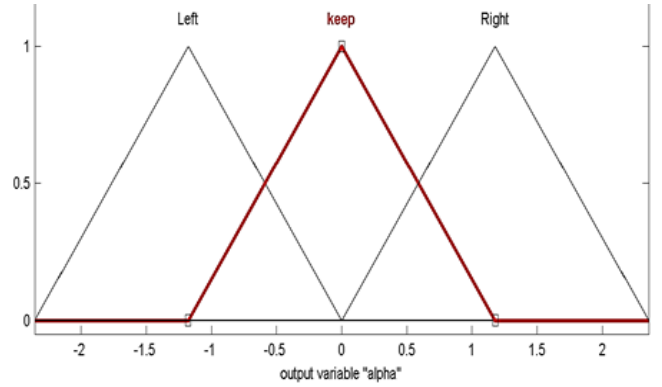


Fig. 23 the results of compensation

D. Presence of obstacles in our control approach

With installation of Acoustic sensors around the truck region, we can compute the distance from the obstacle. Also with computations we can obtain the angle of obstacle with respect to the vehicle position. Using the following formula angle and distance can be obtained in (8) and (9).

$$R = \sqrt{(x_b - X_T)^2 + (y_b - Y_T)^2} \quad (8)$$

$$\mu = \arcsin\left(\frac{x_b - X_T}{y_b - Y_T}\right) - \gamma \quad (9)$$

In equation (8) the variable and are horizontal and vertical distance from the obstacle and with using it we can compute the obstacle distance form truck. And in (7) we can compute the angle between the obstacle and truck. The inference engine in all of those is mamdani and we used the center of surface for difuzzification and fuzzy rules are as follows.

If (μ is Negative & R is Very Near) then (γ is Negative)

If (μ is Positive & R is Very Near) then (γ is Positive)

This controller is parallel to the first level and second level, and its resulting sum is added to second level. And also membership function for angle in presence of obstacle is plotted in Fig. 24.and distance from obstacle and the fuzzy rules for changing the angles in presence of obstacle are shown in Fig. 25 and Fig. 26 respectively. And finally we showed the layers of rules in Fig. 27.

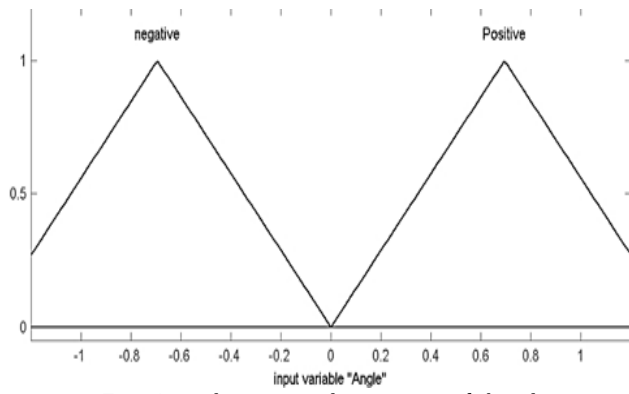


Fig. 24 introduction to angle in presence of obstacle

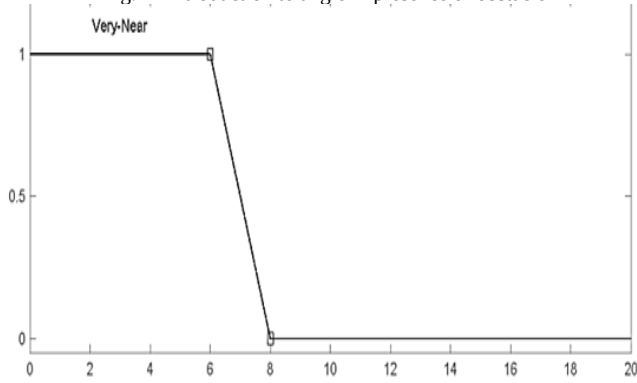


Fig. 25 distance from obstacle

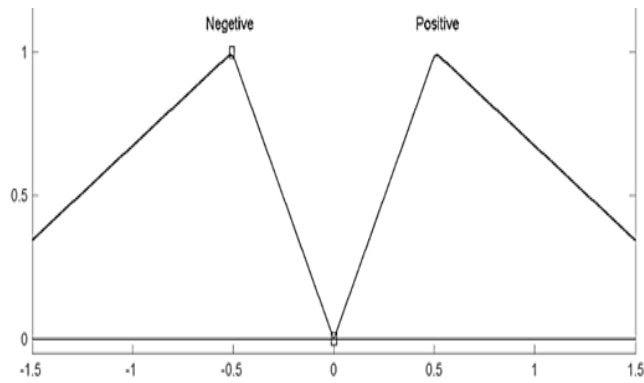


Fig. 26 The result of the changing angle in presence of obstacle

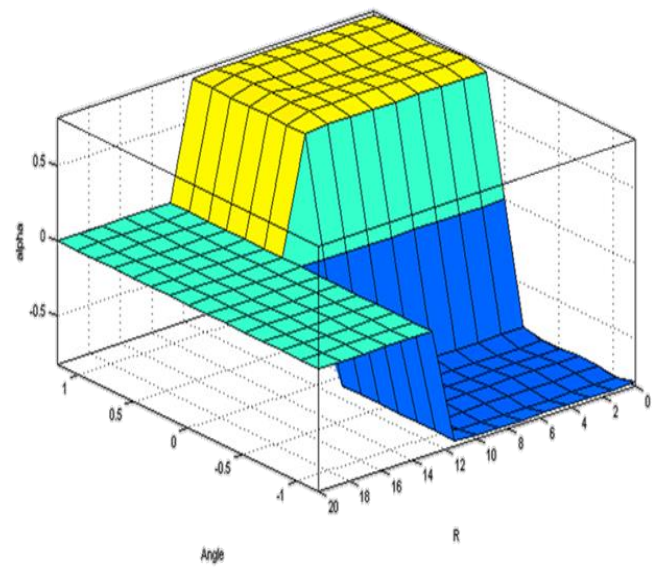


Fig. 27 layers of rules

IV. SIMULATION RESULTS

The simulation results of this approach were surprisingly better from other articles and especially when compared with results obtained by other control strategies and also it should be said that in this approach we don't use any optimization method and final simulation is plotted in Fig. 28 and Fig. 29 for two different positions of obstacle in the path of the vehicle.

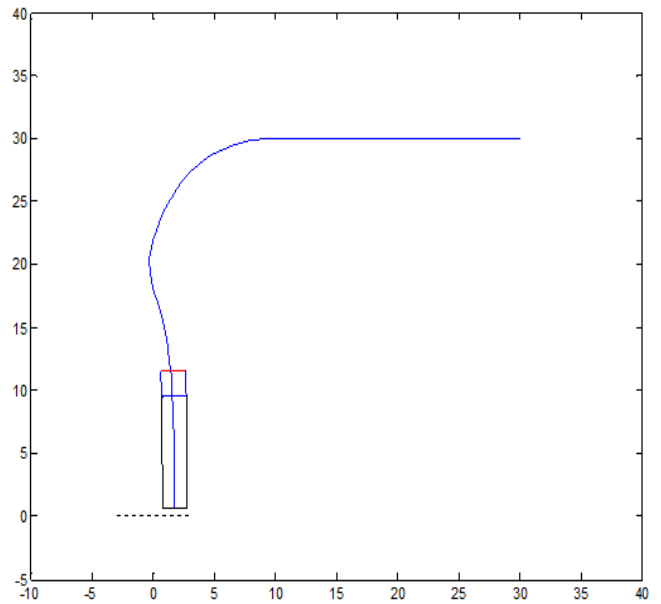


Fig. 28 The result of blockage in the horizontal direction

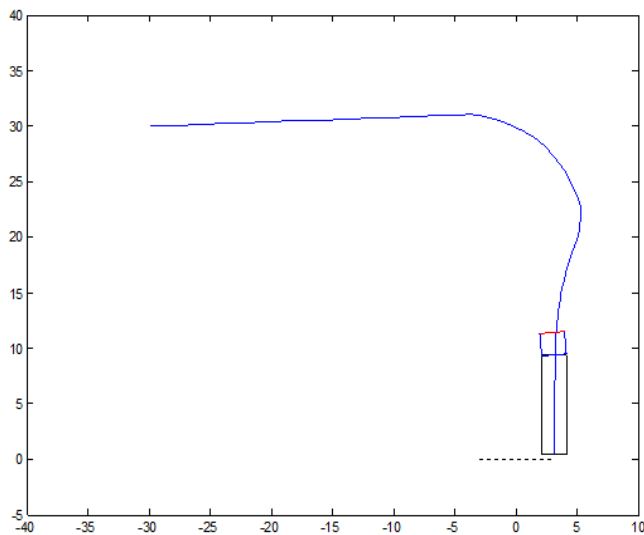


Fig. 29 Simulation of an obstacle in a vertical direction

II. CONCLUSION

In this paper, a fuzzy logic solution for the classical problem of reverse parking problem of trailer-truck was proposed. The main problem is split to Simple solutions .the objective is not to reduce the inputs or etc. while proposing the simple method with reduced number of fuzzy rules. In The presented method the vehicle can pass the obstacles in parking lot with increasing the steering angle of vehicle around the obstacles Which only requires a acoustic sensor and does not need to Metering and expensive cameras and image processing with heavily processing resource. This method is used the simple mathematical and few need parameters and this approach can be done in more difficult and other situations.

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